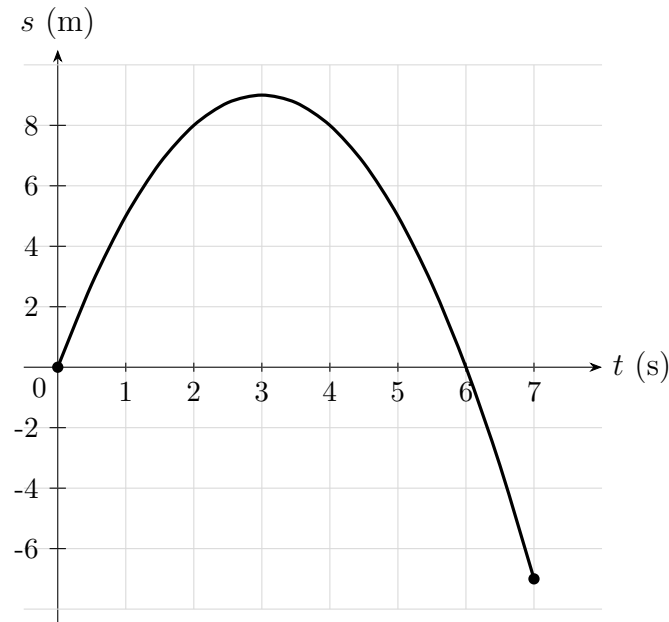


5.4 Classwork: Displacement and Distance from Velocity Curves

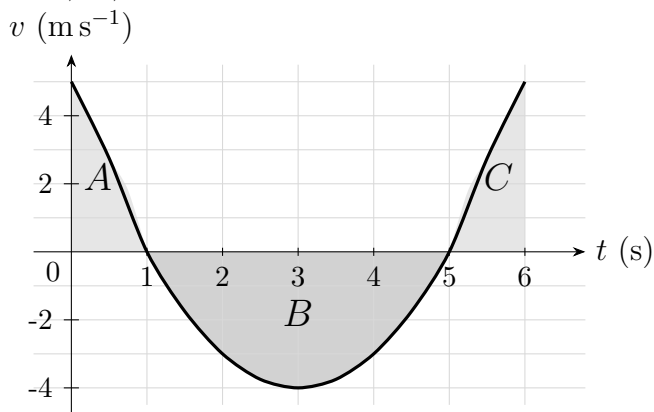
1. The displacement s (metres) of a particle from the origin is shown for $0 \leq t \leq 7$.



- (a) Write down the initial displacement of the particle. [1]
- (b) At what time is the displacement greatest? Write down the maximum displacement, rounded to the nearest metre. [2]
- (c) At what time is the velocity of the particle equal to zero? Explain how you can tell from the graph. [2]
- (d) At what time does the particle return to the origin? [1]
- (e) Is the particle moving toward or away from the origin at $t = 5$? Explain your reasoning. [2]

2. A particle moves along a straight line with velocity $v(t) = t^2 - 6t + 5 \text{ m s}^{-1}$, for $0 \leq t \leq 6$.

The graph of v against t is shown below. The shaded regions between the curve and the t -axis have areas A , B , and C .



- (a) Write down the times when the particle is at rest. [1]
- (b) State the time interval during which the particle moves in the negative direction. [1]
- (c) Using the labeled areas, write down an expression for:
- the displacement of the particle from $t = 0$ to $t = 6$ [1]
 - the total distance travelled from $t = 0$ to $t = 6$ [1]
- (d) Find $\int_0^6 v(t) dt$. [3]
- (e) Find $\int_1^5 v(t) dt$. [2]
- (f) Hence find the total distance travelled from $t = 0$ to $t = 6$. [3]
3. The velocity of a particle is $v(t) \text{ m s}^{-1}$ for $0 \leq t \leq 8$. The particle changes direction when $t = 5$, and $v(t) > 0$ for $0 \leq t < 5$.
- It is known that $\int_0^5 v(t) dt = 12$ and $\int_5^8 v(t) dt = -5$.
- Find the displacement of the particle from $t = 0$ to $t = 8$. [2]
 - Find the total distance travelled from $t = 0$ to $t = 8$. [2]